

Pensana PLC: A Comprehensive Investment Analysis in the Rare Earth Sector

Executive Summary

Pensana PLC (LSE: PRE) is a UK-listed rare earth exploration and development company strategically positioned to address the burgeoning demand for magnet metals from the electric vehicle (EV) and offshore wind industries. The company's core strategy revolves around establishing a sustainable and independent supply chain outside of China, leveraging its flagship Longonjo NdPr Project in Angola and the Saltend rare earth refinery in the UK.

Pensana's valuation is primarily underpinned by the Longonjo project, which, despite a recent re-baselining of economic projections, maintains a significant post-tax Net Present Value (NPV). The market's assessment of Pensana is increasingly influenced by the tangible de-risking milestones achieved at Longonjo, including securing financing and advancing construction. Complementing this, the early-stage Coola exploration project is gaining recognition for its substantial long-term potential, offering a strategic hedge and future upside.

The Longonjo project, characterized by a high-grade, open-pit carbonatite deposit, benefits from exceptional existing infrastructure, including direct rail access to a deep-water port and access to low-cost hydro-electric power. It is designed to produce a high-value Mixed Rare Earth Carbonate (MREC) for further processing at the Saltend refinery. Construction is progressing on schedule and within budget, with initial production anticipated in late 2026.

The Coola project, an early-stage exploration asset, exhibits geological similarities to Longonjo, suggesting significant potential for additional rare earth mineralization. This positions Coola as a crucial long-term value driver and a strategic counterbalance to the inherent development risks associated with the primary project.

Pensana offers a distinctive investment proposition, combining the de-risking trajectory of a near-term producer with the significant exploration upside typically associated with a junior explorer. This dual approach is further bolstered by the company's robust commitment to sustainable practices and its pivotal role in diversifying the global rare earth supply chain. Key upcoming catalysts include the continued advancement and successful commissioning of the Longonjo project, the release of further definitive drill assays and a potential maiden Mineral Resource Estimate (MRE) for Coola, and the securing of additional strategic offtake agreements.

I. Introduction & Current Valuation

Pensana PLC is a London Stock Exchange-listed company focused on the exploration and

development of rare earth elements, particularly neodymium (Nd) and praseodymium (Pr), which are critical for permanent magnets used in electric vehicles and offshore wind turbines. The company's strategic vision emphasizes the creation of an independent and sustainable magnet metal supply chain, aiming to meet a projected 350% increase in magnet metal demand from EVs in the next five years and a 1500% increase from offshore wind over the next two decades.¹

Current Market Data

As of July 29, 2025, Pensana PLC (PRE:LSE) closed at a trading price of 71.20 GBX.² The market capitalization for Pensana PLC on the same date was £225.71 million.²

To provide the market capitalization in USD, an exchange rate conversion is necessary. As a current GBP/USD exchange rate is not available within the provided data, a hypothetical rate of 1.25 USD per 1 GBP will be applied for illustrative purposes. Based on this assumption, the market capitalization in USD would be approximately US\$282.14 million (£225.71 million * 1.25). It is important to note that this USD figure will fluctuate with real-time currency exchange rates.

Valuation Metric (e.g., P/NPV)

Pensana's primary project, Longonjo, has been subject to detailed economic assessments, providing key valuation metrics. An earlier Front-End Engineering Design (FEED) study from May 2022 indicated an unleveraged post-tax NPV8% of US\$3.5 billion.⁴ This valuation was based on an NdPr oxide price assumption of \$153/kg in 2022, with a Compound Annual Growth Rate (CAGR) of 3.7% per annum until 2035, remaining flat in real terms thereafter.⁴

A more recent corporate presentation from March 2024 provides an updated forecast, indicating an NPV@8% of US

1,116 million for the combined Stage 1 and Stage 2 of the Longonjo project. This updated calculation uses an NdPr price assumption of \$150/kg, kept flat in real terms thereafter. The substantial difference between the earlier US\$3.5 billion NPV and the more recent US

1,116 billion NPV, despite similar initial NdPr price points, stems from a critical re-baselining of the project's financial model. The shift from an escalating NdPr price assumption to a flat real price after 2035, coupled with likely updated cost estimates and project phasing, contributes to this reduction. This re-baselining suggests a move towards a more conservative or updated economic outlook for the project, which can be interpreted as a step towards de-risking by setting more achievable financial targets.

Using the latest market capitalization (£225.71 million as of July 29, 2025²) and the most recent NPV8% for Longonjo (US

1,116 million as of March 2024⁷), and applying the hypothetical GBP/USD exchange rate of 1.25, the calculated Price – to – Net Present Value (P/NPV) for Pensana is approximately 0.25x (US

282.14 million / US\$1,116 million). The NdPr price assumption for this NPV calculation is \$150/kg, kept flat in real terms thereafter.⁷

A P/NPV ratio of 0.25x is significantly below 1.0x, indicating that the market is currently valuing Pensana at a substantial discount relative to the stated net present value of its primary project. This "development discount" is a common phenomenon for mining companies during the construction phase. It reflects several factors, including the perceived execution risk

associated with bringing a project to production (e.g., potential for construction delays, cost overruns, or operational ramp-up challenges), skepticism regarding the long-term sustainability of future NdPr price forecasts, or simply the time value of money until the project begins generating consistent cash flows. This low P/NPV suggests a considerable potential for undervaluation if the company successfully executes its development plan, or it could signal that the market perceives high inherent risks that must be overcome before a re-rating occurs.

Table 1: Pensana Current Market Data & Valuation Snapshot

Metric	Value	Date/Assumption	Source
Current Share Price (GBX)	71.20 GBX	July 29, 2025	²
Market Capitalization (GBP)	£225.71 million	July 29, 2025	²
Market Capitalization (USD)	US\$282.14 million	July 29, 2025 (Hypothetical 1.25 GBP/USD)	Calculated
Longonjo NPV8% (Post-tax, unleveraged)	US\$1,116 million	March 2024 Presentation (Stage 1 & 2)	⁷
NdPr Price Assumption for NPV	\$150/kg, flat real thereafter	March 2024 Presentation	⁷
Calculated P/NPV	0.25x	July 29, 2025 / March 2024 NPV	Calculated

This table offers a clear, concise overview of Pensana's current market standing relative to its flagship project's intrinsic value. By explicitly stating the key assumptions, such as the NdPr price and the hypothetical exchange rate, it provides transparency crucial for financial analysis and allows for further independent validation or sensitivity analysis. The P/NPV metric visually highlights the current market discount, setting the groundwork for a deeper exploration into the factors driving this valuation gap.

II. Key Value Drivers & Market Perception

Pensana's market perception and stock performance are intricately linked to its progress in developing its key rare earth projects, particularly Longonjo, and the broader market dynamics for critical minerals.

Recent Price Drivers

Pensana's share price has exhibited notable volatility over recent months, as demonstrated by significant daily fluctuations in July 2025, including a +12.15% increase on July 8 and a -5.07% drop on July 29.⁸ A primary driver of this volatility has been updates concerning the Longonjo project. For instance, the announcement on July 28, 2025, confirming that "Longonjo construction budget on track and Coola carbonatite drilling" was a recent and potentially impactful event.² Despite this seemingly positive news regarding construction progress, the share price experienced a 9.2% decline to 74.80 pence on Monday afternoon, July 28, 2025.¹⁰

This observed disconnect between positive project news and an immediate negative stock price reaction warrants closer examination. While confirming that construction is on schedule and within budget, and initiating drilling at Coola, are fundamentally de-risking developments for the primary project and unlock upside potential for the secondary, the market's response suggests several possibilities. It is plausible that these expectations were already largely priced into the stock, or that broader market factors, such as a general downturn in the junior mining sector or profit-taking after previous surges, overshadowed the specific company news. The recent equity issuance in June 2025 for employee incentive programs might also have contributed to selling pressure.¹² This scenario indicates that merely meeting market expectations may not be sufficient for a significant re-rating; exceeding them or demonstrating sustained operational success and cash flow generation is often required to trigger a substantial positive price movement.

Other positive developments that have influenced the stock include the Memorandum of Understanding (MOU) with ReElement Technologies in June 2025, aimed at developing a sustainable rare earth supply chain and a proposed offtake of up to 20,000 tonnes per annum (tpa) of Mixed Rare Earth Carbonate (MREC) from Longonjo.¹ A similar MOU with Toyota Tsusho Corporation for rare earth offtake was announced on June 15, 2025.³ A more significant price surge, a 29% increase to 36.00 pence, was observed on March 18, 2025, following the announcement of secured full financing approvals for the Longonjo project, amounting to approximately US\$268 million.²⁰

Specific Projects/Deposits Significantly Influencing Market Perception

The **Longonjo Project** remains the dominant factor shaping market perception. Its tangible progress towards construction, the successful securing of comprehensive financing¹⁰, and the establishment of key offtake agreements¹³ are critical in de-risking the project and signaling its transition from a developer to a future producer. The consistent confirmation that the US\$217 million capital cost envelope remains intact is a key element in building and maintaining investor confidence.¹⁰

The **Saltend Refinery** in the UK also contributes significantly to market perception. Its strategic role in establishing a non-Chinese rare earth supply chain, coupled with backing from the UK government's Automotive Transformation Fund, underscores its importance in the context of increasing geopolitical focus on critical mineral supply security.¹

While in an earlier stage of development, the **Coola Exploration Project** is beginning to influence market perception as a significant long-term growth and upside factor. The commencement of drilling at Coola¹⁰ and its stated potential to extend the mine life of Longonjo's processing infrastructure beyond 20 years¹⁰ signals future resource potential and operational longevity.

Market's Valuation Attribution

The market is currently attributing value to Pensana primarily through its **near-term production potential** from the Longonjo project. The visible progress in construction, the successful securing of financing, and the establishment of crucial offtake agreements are all vital indicators of this potential. The company's strategic decision to produce Mixed Rare Earth Carbonate (MREC) as a midstream product¹⁵, which is then planned for further processing at the Saltend refinery, adds a layer of value-add that is recognized by the market as enhancing the overall value chain.

The **large resource size** of Longonjo also underpins the long-term intrinsic value. It is consistently described as "one of the world's largest and highest-grade magnet metal rare earth deposits".⁵ With a JORC Reserve of 139,457 tonnes of NdPr oxide (NdPrO)⁷ and a projected mine life exceeding 20 years⁷, the sheer scale of the deposit provides a strong foundation for future production.

Furthermore, the market assigns value to Pensana's **strategic importance and strong Environmental, Social, and Governance (ESG) credentials**. Its unique position in developing a non-Chinese, sustainable supply chain for critical magnet metals is a key differentiator in the current global geopolitical landscape.¹ This commitment to sustainability, including the use of hydropower and electrified infrastructure, aligns with increasing investor and consumer demand for ethically sourced materials.

For junior mining companies, the basis of valuation typically evolves as a project matures. Early-stage exploration companies are often valued based on the "pounds in the ground" or the sheer size of their mineral resources, along with the speculative upside of new discoveries. As a company progresses through feasibility studies, permitting, financing, and moves into construction, the valuation methodology shifts. It transitions towards discounted cash flow (DCF) models, which place a greater emphasis on the project's economics, including Net Present Value (NPV), Internal Rate of Return (IRR), and projected cash flows, as well as the increasing probability of achieving commercial production. Pensana is clearly in this transitional phase. The market is increasingly looking for tangible progress on construction, successful financing, and confirmed offtake agreements, signaling a move from pure exploration speculation to a more cash-flow-driven valuation. The low P/NPV discussed previously

indicates that while the market acknowledges the project's substantial potential, it is still applying a significant discount due to the inherent development and execution risks. Pensana's future valuation will therefore depend increasingly on its ability to successfully execute the Longonjo project, meet production targets, and demonstrate consistent cash flows, rather than solely on further resource upgrades.

Baseline Valuation

The announcement concerning "Longonjo construction budget on track and Coola carbonatite drilling" was released on July 28, 2025.² To establish a baseline valuation immediately prior to this news, the market capitalization on July 25, 2025, the last trading day before the weekend and Monday's announcement, can be used as a proxy. On July 25, 2025, Pensana's market capitalization was £243.53 million.³²

Applying the hypothetical GBP/USD exchange rate of 1.25, this translates to approximately US *304.41million*. *Using the March 2024 NPV of US* ^{1,116} million for Longonjo ⁷, the P/NPV immediately prior to the announcement would have been approximately 0.27x (US *304.41million/US* ^{1,116} million). This represents the market's valuation of the Longonjo project's potential at that specific point, prior to the latest updates on construction and Coola drilling.

Unrealized Value

Given the recent information, the full value of the Longonjo project has likely not yet been fully captured by the market. The persistent low P/NPV ratio, even after positive updates, suggests that a significant "development discount" remains. While the market acknowledges progress in construction and financing, it appears to be waiting for more definitive de-risking milestones.

The expected next catalyst for its valuation is the successful commissioning and commencement of initial production at the Longonjo mine, anticipated in late 2026.¹⁸ Demonstrating consistent production of high-value Mixed Rare Earth Carbonate (MREC) and achieving projected cash flows will be crucial. Further catalysts include the successful ramp-up to Phase 2 production, which would double capacity to 40,000 tpa of MREC ²³, and the successful integration of Longonjo's MREC with the Saltend refinery, demonstrating a fully operational mine-to-magnet supply chain. Additionally, positive drill results and a potential maiden Mineral Resource Estimate (MRE) from the Coola exploration project could unlock further long-term value, as discussed in subsequent sections.

III. Analysis of Primary Project: Longonjo

Pensana's primary near-term project is the Longonjo Rare Earths Project, located in Angola. This project is central to the company's strategy of establishing an independent and sustainable rare earth supply chain.

Project Name & Type

The primary near-term project is the **Longonjo Rare Earths Project**, situated in the Huambo Province of central Angola.¹⁷ It is characterized as an **open-pit** mining operation, predominantly involving "free-dig" material.¹⁷ Geologically, the deposit is a **weathered carbonatite** formation, between 20 and 30 meters thick, containing significant concentrations of monazite and bastnaesite, which are key rare earth minerals.¹⁷

Processing Method & Advantages

The processing method for Longonjo involves extracting, concentrating, calcining, and chemically refining the free-dig material to produce a **high-value Mixed Rare Earth Carbonate (MREC)**.¹⁷ This MREC is considered a midstream product, representing a significant step up the value chain compared to the basic concentrates typically exported from developing nations.²⁸ The MREC product is described as "super clean" and has been laboratory-tested and validated with potential customers, confirming its marketability.²⁸

Key advantages of this processing method and the overall project design include:

- **Cost Efficiency:** The project benefits from existing infrastructure and low-cost hydro-electric power, which is contracted at US\$0.02/kWh for 10 years.⁵ The electrification of traditionally energy-intensive plant items, such as the kiln at Longonjo, is projected to save 25,000 tonnes of CO₂ equivalent per year.²⁴
- **Environmental Impact:** Pensana is committed to developing the Longonjo mine with the lowest carbon footprint, utilizing low-carbon solar energy and hydropower.¹⁷ The project design incorporates reagent recycling and is located near existing infrastructure, reducing the need for diesel-intensive road transport.²⁴ The tailings storage facility is designed to meet the Global Industry Standard on Tailings Management (GISTM).¹⁷ The overall approach is expected to cut over 4 million tonnes of CO₂ by replacing fossil fuel electricity with hydro-powered operations.²²
- **Product Quality & Marketability:** Producing MREC rather than a raw concentrate creates a higher-purity and higher-value product with a broader market, not limited solely to China.³³ This allows Pensana to establish an independent, sustainable supply of magnet metal oxides, catering to a market currently dominated by China.²⁴ The Saltend refinery, designed to process this MREC, will produce separated rare earth oxides, including 4,500-5,000 tpa of NdPr oxide, representing over 5% of the world market in 2025.⁵

Infrastructure

The Longonjo project boasts critical infrastructure that significantly enhances its viability and cost-effectiveness. It has convenient access to a sealed national road and, crucially, the **Lobito Corridor railway line**, which passes within approximately 5 km of the project area.¹⁷ This railway provides a direct link for transporting the MREC 273 km to the **Atlantic deep-water**

port of Lobito for export.¹⁷

For power supply, the mine is strategically situated near the national grid, which is linked to the **2GW Laúca hydropower scheme**. This provides access to very low-cost hydro-electric power, a key advantage for reducing operational expenses and environmental footprint.⁵ The project also includes provisions for a process water supply and comprehensive mine infrastructure, workshops, offices, and an accommodation village.¹⁷ This existing and planned infrastructure greatly reduces the need for substantial greenfield development costs typically associated with remote mining projects.

Capital Expenditure (Capex) & Cost Comparison

The initial capital expenditure (capex) for the Longonjo project is US

217million.[7, 10, 11, 23, 31] This figure includes contingency and is considered among the lowest for a rare earths project of its scale, largely due to the existing infrastructure and low strip ratio.[31, 36] A portion of this capex, approximately US

105 million, is deferred until year three, which helps manage initial financing requirements.⁷

While direct industry averages for "capex per ton of average production" specifically for rare earths brownfield and greenfield sites are not explicitly provided in the snippets, comparisons can be drawn from other rare earth projects mentioned. For instance, a general rare earth mining and refining model suggests a capex of

150,000/Tpa (tonnes per annum) to achieve a 20 421 million for its Ngualla project, which is expected to generate average annual EBITDA of US

448million.[38] Pensana's US 217 million capex for an initial 20,000 tpa MREC

production (scalable to 40,000 tpa)²³ appears competitive, particularly given the high-value MREC product and the significant NdPr content. The low capex is explicitly attributed to the excellent existing rail, hydro, and water infrastructure.⁷ This efficiency in capital deployment is a significant advantage, as it reduces the financial hurdle for project development and enhances its overall economic viability.

Economic Indicators

The Longonjo project demonstrates robust economic indicators, as outlined in the March 2024 corporate presentation.⁷ Key metrics include:

- **Internal Rate of Return (IRR):** 34% (for Stage 1) and 29% (for Stage 2).⁷ An earlier FEED study (May 2022) reported an IRR of 71%⁴, while a March 2022 operational update indicated 61%.³⁵ The most recent figures reflect a re-baselined, potentially more conservative, but still attractive return profile.
- **Payback Period:** 4 years for both Stage 1 and Stage 2.⁷ An earlier estimate was 1.5 years.⁴
- **Estimated Payable NdPr Content:** The Longonjo project has a JORC compliant reserve of 21.54 million tonnes at 3.04% Total Rare Earth Oxide (TREO), containing 139,457 tonnes of NdPr.¹⁷ The initial production phase is targeted to produce 20,000 tpa of Mixed Rare Earth Carbonate (MREC)¹⁵, which can be scaled up to 40,000 tpa in a proposed second

phase.²³ This output is projected to represent approximately 5% of the world's magnet metal rare earths production.¹⁸

The economics of rare earth projects, including Longonjo, are sensitive to NdPr market prices. The spot price for NdPr on the Shanghai Metals Market (SMM) platform had languished below \$50,000/MT but started increasing six weeks prior to September 6, 2025, reaching \$61,357/MT.³⁹ This price recovery is a positive development for the project's revenue outlook.

Risk & Probability

The risk level and probability of the Longonjo mine being built appear to be significantly de-risked and high, respectively, given recent developments. Pensana has successfully secured full financing approvals for the project, totaling approximately US

268million.^[20, 21, 22] *This includes a US* 160 million syndicated loan facility from the Africa Finance Corporation (AFC) and Absa Bank, along with equity and convertible loan investments from the Angolan Sovereign Wealth Fund (FSDEA) and AFC.²⁰ The FSDEA has also provided a US\$15 million bridging loan, ensuring project momentum.⁷

Furthermore, construction on the Longonjo mine commenced in May 2025, and as of July 28, 2025, the project team confirmed that the US\$217 million capital cost envelope "remains intact" and that the project is on schedule.¹⁰ Key project delivery partners are mobilized and engaged in detailed engineering activities.¹⁰ Offtake agreements, such as the five-year agreement with Toyota Tsusho for the full initial yearly production, and an MOU with ReElement Technologies for up to 20,000 tpa of MREC, further reduce market risk.¹¹ The project is also fully licensed and permitted to operate, with mining rights, environmental and social impact assessments (ESIAs), and environmental licenses in place.¹⁷ These factors collectively indicate a high probability of the mine being built and reaching production.

Next Major Driver

The next major driver or milestone expected for the Longonjo project is the **successful commissioning and commencement of initial production**, which is anticipated in late 2026.¹⁸ This transition from construction to operational status will be a pivotal moment, validating the project's technical and financial viability and initiating cash flow generation. Further significant milestones include the ramp-up to full Phase 1 production capacity (20,000 tpa MREC) and the subsequent expansion to Phase 2 (40,000 tpa MREC).¹⁸ Continued progress on the Saltend refinery in the UK, which will process the MREC, is also a critical complementary driver.

Valuation Focus

The value of the Longonjo project primarily lies in its **economics and near-term production potential**, rather than solely its intrinsic resource size. While the large resource size (138 million tonnes at 2.19% TREO and 0.47% NdPr in Mineral Resources, with 21.54 million tonnes at 3.04% TREO and 0.65% NdPr in Mineral Reserves⁵) provides a robust foundation and a long mine life

(20+ years ⁷), the market's current valuation is heavily influenced by the project's ability to translate these resources into profitable cash flows in the near future.

The emphasis on economic indicators like IRR (34-29% ⁷) and the relatively low capital expenditure for its scale ⁷ are key factors. The transition from an exploration-stage company, typically valued on resource size, to a development-stage company, where project economics and execution become paramount, is evident. The market is now scrutinizing the project's ability to meet its production targets, manage costs, and secure favorable pricing for its MREC. The confirmed financing and construction progress are direct indicators of this near-term potential, making the economics and the pathway to production the primary drivers of value.

Critical Assumptions/Sensitivities

The critical assumptions and potential 'make or break' factors for Longonjo's economics include:

- **NdPr Oxide Price:** The project's financial viability is highly sensitive to the price of NdPr oxide. The latest NPV calculation assumes a flat real price of \$150/kg after 2035.⁷ Any significant deviation from this forecast, particularly a sustained decline, could materially impact revenue and profitability.
- **Capital Expenditure (Capex) Numbers:** While the US\$217 million capital cost envelope has been confirmed as intact ¹⁰, any unforeseen cost overruns during construction or ramp-up could negatively affect the project's economics and require additional financing.
- **Deposit Type Consistency and Processing Method Feasibility:** The project relies on the consistency of the weathered carbonatite deposit and the successful application of the chosen processing method to produce a high-value MREC.¹⁷ While pilot-scale testing has been conducted ²⁸, scaling up to commercial production always carries inherent technical risks. The ability to produce an "ultra-clean Mixed Rare Earth Carbonate" that meets customer specifications is crucial for market acceptance and pricing.¹⁴
- **Offtake Agreements:** The ability to secure and maintain long-term offtake agreements, such as those with Toyota Tsusho and ReElement Technologies, is vital for ensuring market access and stable revenue streams.¹¹
- **Operational Ramp-up:** The successful and timely ramp-up to initial production and then to Phase 2 capacity is critical for achieving projected cash flows and payback periods. Delays or operational inefficiencies could impact profitability.

IV. Analysis of Secondary/Exploration Project: Coola

Beyond its flagship Longonjo project, Pensana holds significant exploration upside in its secondary project, Coola, which includes the Sulima West target. This project is strategically important for the company's long-term growth and risk mitigation.

Project Name & Role

Pensana's secondary or exploration project is the **Coola Exploration Project License**, which

also encompasses the **Sulima West** target area.⁴⁰ Located in central Angola, approximately 40 to 100 kilometers north of the Longonjo project, Coola plays a crucial role as a **strategic 'hedge' or support** for the overall company value.⁴⁰

Its primary role is to provide potential additional high-grade rare earth feedstock that could extend the operational life and optimize the utilization of the processing infrastructure being developed at Longonjo.¹⁰ By identifying and delineating new resources within trucking distance, Coola can significantly increase the overall production potential and longevity of Pensana's Angolan operations, acting as a long-term growth engine beyond the initial 20-year mine life of Longonjo.

Current Market Valuation & Hype

The market's current perception and attributed value to the Coola project are nascent but growing. Given its early exploration stage, a definitive numerical NPV is not yet available, and its value is largely speculative, driven by geological prospectivity and the potential for significant discoveries. There is considerable "hype" surrounding Coola due to its geological similarities to Longonjo and the potential for extremely large amounts of rare earth elements, particularly NdPr, to be extracted.

Pensana's Exploration Manager, Grant Hayward, has stated a belief that Coola is a "highly prospective volcanic pipe which appears to have a similar scale to Longonjo".⁷ Early soil sampling programs at the Coola carbonatite have identified high tenor soil anomalies up to 4.69% Total Rare Earth Oxide (REO) extending over a 1.3 x 1.4 kilometer area.⁴³ Additionally, trench and pit sampling results at Sulima West have confirmed a six-hectare enriched laterite grading 2-3% TREO, with higher-grade zones up to 10.6% TREO in initial reconnaissance samples.⁴⁰ These early indications suggest the potential for substantial rare earth mineralization, fueling market interest in its long-term resource potential.

Geological Style

The geological style of mineralization for the Coola project is primarily associated with **carbonatite and alkaline complexes**, similar to Longonjo.⁴³ The Coola carbonatite itself is described as a roughly circular, 900-meter wide body comprising an incomplete carbonatite ring dyke surrounded by fenite, enclosing a soil-covered central diatreme of approximately 600 meters in diameter.⁴⁰ This structure is known to be enriched in elements and minerals associated with rare earth mineralization, such as fluorite, strontium, barite, phosphorus, and niobium.⁴³ The presence of a compelling magnetic anomaly in the central diatreme suggests potential for deeply weathered and supergene enriched rare earth mineralization, mirroring the style found at Longonjo.¹⁰

At **Sulima West**, the mineralization is found in an **alkaline complex with rare earth element and phosphate (P₂O₅) enriched regolith**, specifically within an iron/manganese-rich laterite that is very similar in appearance to the rare earth element laterite developed over the Longonjo carbonatite.⁴⁰ Preliminary mineral liberation studies indicate that the Rare Earth

Element (REE) bearing phases are monazite, bastnaesite, and florencite, showing moderate liberation and exposure.⁴⁰

Market Context (Supply Gap)

The potential amount of rare earth elements, particularly NdPr, from the Coola project fits directly into the broader market context of an imminent supply gap for these critical metals. The global demand for magnet metals is projected to increase significantly, driven by the rapid expansion of electric vehicles and offshore wind turbines.¹ The International Energy Agency (IEA) suggests the world will need to increase rare earth supply by over 700% by 2040 to meet renewable technology demand.²⁴

China currently dominates the rare earth supply chain, with Europe depending on China for 98% of its rare earth magnets.²⁵ Pensana's strategy, including the development of Longonjo and exploration at Coola, aims to establish an independent, sustainable, and non-Chinese supply chain.¹⁵ If Coola proves to host significant additional NdPr resources, it would further bolster Pensana's ability to contribute to this critical supply diversification, helping to alleviate the projected global supply deficit and enhance supply chain resilience for Western industries. The potential for Coola to extend the mine life of Longonjo's processing infrastructure beyond 20 years directly addresses the long lead times for new mines and the associated infrastructure, which are contributing to the risk of supply bottlenecks.⁷

Challenges & Stage

The Coola project, being an early-stage exploration asset, faces several inherent risks and challenges:

- **Early Stage:** Coola is currently in the exploration phase, with initial drilling programs underway to confirm resource potential.¹⁰ While initial soil and trench sampling results are encouraging⁴⁰, these are not yet sufficient to define a JORC-compliant Mineral Resource Estimate (MRE).
- **Lack of Definitive Resource Estimates:** The project lacks clear resource estimates or detailed economic studies (like a Preliminary Economic Assessment or Feasibility Study) that would allow for a precise numerical Net Present Value (NPV) calculation.⁴⁶ Valuation at this stage is typically based on comparative metrics, geological potential, and exploration success.
- **Need for New Mine Build (Potentially):** While the current plan is to truck Coola concentrates to the Longonjo facility for processing⁴⁰, if Coola proves to be a standalone, exceptionally large deposit, it might eventually necessitate a new, dedicated mine build, which would involve significant additional capital expenditure and development timelines.
- **Exploration Risk:** All exploration projects carry inherent geological and technical risks. The initial drilling program at Coola is designed to confirm resource potential and allow for follow-up drilling.¹⁰ The success of these programs is not guaranteed, and the

ultimate economic viability of the mineralization remains to be proven.

Due to these factors, assigning a numerical NPV at Coola's current stage is difficult. Valuation methods for early-stage projects often rely on metrics such as Enterprise Value (EV) per unit of contained resource, benchmarking against comparable discoveries, or qualitative assessments of geological prospectivity.⁴⁸ The focus remains on de-risking the exploration process through systematic drilling and metallurgical test work to define a resource that can eventually be incorporated into a formal economic assessment.

Long-term vs. Short-term Value

The Coola project represents a **long-term play** for Pensana. Its current stage of exploration means it is not expected to be a short-to-mid-term catalyst for immediate production or cash flow. Instead, its value lies in its potential to significantly extend the mine life of Longonjo's processing infrastructure beyond the currently contemplated 20 years.¹⁰ This longevity provides a sustained supply of critical rare earths, reinforcing Pensana's position in the global market for decades to come.

Coola acts as a '**shield against downside risk**' from the Longonjo project in several ways. Firstly, by offering a potential future source of high-grade feedstock, it mitigates the risk of resource depletion at Longonjo over the very long term. Secondly, it provides optionality and flexibility in Pensana's overall rare earth supply strategy. If Longonjo faces unexpected operational challenges or its resource proves less extensive than currently estimated, Coola could step in as a supplementary or even primary source of material. This geological diversification reduces the company's reliance on a single deposit, enhancing its overall resilience and long-term asset base. The ongoing exploration at Coola, and the promising early results, contribute to the company's overall asset value and future growth prospects, providing a layer of security against unforeseen issues at the primary project.

V. Valuation of Secondary/Exploration Project (if applicable)

Valuing an early-stage exploration project like Coola, which lacks clear resource estimates or detailed economics, presents a significant challenge. Traditional discounted cash flow (DCF) models are not applicable due to the high level of geological and economic uncertainty.

Valuation Approach

Given Coola's early stage, the most appropriate valuation approach involves **comparative metrics** and **qualitative assessments of geological potential**. The Enterprise Value (EV) to Resource ratio is a common metric used for early-stage development projects where detailed information for a DCF analysis is not yet available.⁴⁸ This involves comparing the company's enterprise value to the total resources identified in the ground.

For Coola, the valuation currently relies on:

- **Geological Analogies:** Its similarities to the Longonjo carbonatite in terms of geological style and mineralization ⁴¹ provide a qualitative benchmark.
- **Exploration Results:** Promising early-stage drill results, soil anomalies, and trench sampling results, such as high tenor soil anomalies up to 4.69% REO at Coola carbonatite and 2-3% TREO at Sulima West ⁴⁰, serve as indicators of potential scale and grade.
- **Strategic Value:** Its potential to extend the mine life of Longonjo's processing infrastructure and contribute to a non-Chinese rare earth supply chain adds strategic value that is difficult to quantify numerically at this stage.

Estimated Resource Size

While a JORC-compliant MRE is not yet available for Coola, preliminary indications suggest significant potential. Pensana's Exploration Manager has stated that Coola appears to have a "similar scale to Longonjo".⁷ Longonjo's JORC Reserve contains 139,457 tonnes of NdPr oxide.⁷ If Coola were indeed to host a comparable amount, it would represent a very substantial potential payable NdPr content.

Early exploration results reinforce this potential:

- Soil sampling at Coola carbonatite identified high tenor soil anomalies up to 4.69% REO over a 1.3 x 1.4 kilometer area.⁴³
- Trench and pit sampling at Sulima West confirmed a six-hectare enriched laterite grading 2-3% TREO, with higher-grade zones up to 10.6% TREO.⁴⁰
- Rock chip sampling of the Coola carbonatite ring dyke returned values between 0.6% and 4.9% TREO (average 2.6%), and soil geochemistry over the covered carbonatite returned values between 0.37% and 13.18% TREO (average 3.21%).⁴²

These results, while not constituting a resource estimate, indicate a strong potential for a large, high-grade rare earth deposit.

Comparison to Primary Project

The potential size of Coola, as suggested by its geological similarities and early exploration results, is considered to be **comparable to Longonjo**.⁷ The Coola carbonatite complex is described as having a "similar size" to Longonjo.⁴¹ Longonjo has a JORC Reserve of 21.54 million tonnes at 3.04% TREO, containing 139,457 tonnes of NdPr ¹⁷, and a total Mineral Resource of 138 million tonnes at 2.19% TREO, containing 650,000 tonnes of NdPr.⁵

If Coola were to delineate resources of a similar magnitude, it would effectively double Pensana's long-term rare earth asset base, providing a significant boost to its overall resource inventory and future production potential. This comparison highlights Coola's role as a potential "company maker" in its own right, beyond just extending Longonjo's mine life.

Risk/Reward Profile

The Coola project embodies a **high risk, high reward** profile.

- **High Risk:** As an early-stage exploration project, it carries inherent geological uncertainty. There is no guarantee that the promising early results will translate into a defined, economically viable resource. The project still requires significant further drilling, metallurgical test work, and economic studies to de-risk it.⁴⁷
- **High Reward:** If Coola successfully delineates a large, high-grade resource comparable to Longonjo, the upside potential for Pensana's valuation is substantial. Such a discovery could significantly increase the company's overall resource base, extend its production profile for decades, and reinforce its strategic position in the global rare earth market. The potential to feed high-grade material to the existing Longonjo processing plant further enhances the reward by leveraging existing infrastructure.¹⁰

Impact of Drill Results/MRE

Drill results and a potential Mineral Resource Estimate (MRE) from Coola are expected to have a **significant positive impact on Pensana's valuation**.

- **De-risking Exploration:** Positive drill results, particularly those confirming continuity and grade, reduce geological risk and increase confidence in the project's potential.⁴⁵ This de-risking typically leads to an increase in attributed value.
- **Resource Definition:** The publication of a maiden MRE would be a transformative milestone. It would convert speculative potential into a quantifiable asset, allowing analysts and investors to incorporate Coola's resources into their valuation models, potentially leading to a re-rating of the company's shares.
- **Future Production Potential:** An MRE would provide the basis for future economic studies (e.g., PEA, PFS), outlining how Coola could contribute to Pensana's long-term production profile, either as a direct feed to Longonjo or potentially a standalone operation. This would enhance the company's long-term investment appeal.
- **Increased Market Attributed Value:** As the project progresses through these stages, the market's implied value attributed to Coola would likely increase, reflecting the reduced risk and clearer path to potential value realization.

Ballpark NPV (if possible)

Given Coola's early stage and the absence of a defined resource estimate or detailed economic studies, providing a precise ballpark NPV is highly speculative and challenging. However, a proportional sizing guesstimate can be considered based on its potential comparison to Longonjo.

Longonjo's NPV8% is US\$1,116 million, based on a JORC Reserve of 139,457 tonnes of payable NdPr oxide.⁷ If Coola were to delineate a resource of similar scale, as suggested by company commentary⁷, its potential NPV could theoretically approach that of Longonjo. However, this is a very high-level guesstimate and would be subject to:

- **Grade and Metallurgy:** The actual payable NdPr content and recovery rates from Coola's mineralization.
- **Mining Method:** While Longonjo is open pit, Coola's optimal mining method (e.g., open pit, underground) would influence capex and opex.
- **Capex and Opex:** The costs associated with developing Coola, even if it feeds into Longonjo's processing plant, would need to be factored in.
- **Discount Rate:** The discount rate applied would reflect the higher risk profile of an exploration project compared to a development project.

Therefore, without a defined resource and preliminary economic assessment, assigning a numerical NPV to Coola would be premature and highly unreliable. Its value is currently best understood as significant exploration upside.

Market's Attributed Value (Quantitative)

Quantifying the implied market value attributed specifically to Coola is complex, as Pensana's market capitalization primarily reflects the Longonjo project's progress. However, stock movements following Coola-specific news can provide an indication. For example, the July 28, 2025, announcement, which included "Coola carbonatite drilling"¹⁰, saw Pensana shares down 9.2%.¹⁰ This suggests that while the news was positive for Coola, it did not immediately translate into a significant positive re-rating for the company as a whole, possibly due to broader market factors or the Longonjo construction update being the primary focus.

The market capitalization of Pensana as of July 29, 2025, was £225.71 million.² If one were to hypothetically subtract the NPV of Longonjo (US\$1,116 million, or approx. £892.8 million at 1.25 GBP/USD) from the market cap, it would result in a negative number, which is illogical. This reinforces that the market is applying a substantial discount to Longonjo's NPV due to development risk. The value attributed to Coola is therefore embedded within this discounted valuation, representing the long-term potential that is yet to be fully priced in.

Comparative Valuation at MRE

Comparing the current implied valuation of Coola (per pound of NdPr, if a guesstimate were possible) to the valuation of Longonjo at its first MRE is not feasible with the provided data, as a specific "first MRE" valuation for Longonjo is not detailed in the snippets. However, the general principle is that early-stage resources, even with an MRE, are valued at a much lower "per pound" metric than resources in advanced development or production. This is due to the higher inherent risks (geological, metallurgical, economic, permitting, financing) that need to be retired as a project progresses.

If Coola achieves a maiden MRE, its implied valuation per pound of NdPr would likely be significantly lower than Longonjo's current implied valuation per pound, reflecting the earlier stage of development and the remaining risks. As Coola progresses through studies and de-risking milestones, its "per pound" value would gradually increase, narrowing the gap with Longonjo. This progression would contribute to Pensana's overall P/NPV by adding a

quantifiable, albeit discounted, long-term asset to its portfolio.

VI. Overall Investment Thesis & Catalysts

Pensana PLC presents a compelling, albeit speculative, investment opportunity in the critical rare earth sector, driven by a dual-project strategy that balances near-term production potential with significant long-term exploration upside.

Preferred Investment Thesis

The strongest investment thesis for Pensana lies in the combination of the **Longonjo project as the near-term producer, complemented by the Coola project as a strategic long-term hedge and growth driver.**

Longonjo provides the foundational value, offering a clear pathway to production of high-value Mixed Rare Earth Carbonate (MREC) by late 2026.¹⁸ Its robust economics, secured financing²⁰, and established offtake agreements¹¹ position it as a de-risked asset moving towards cash flow generation. The Saltend refinery further enhances this thesis by providing a downstream processing hub, creating a vertically integrated, non-Chinese supply chain that is highly attractive in the current geopolitical climate.¹

Coola, on the other hand, acts as a critical long-term value enhancer. While currently an exploration play, its geological similarities to Longonjo and promising early results suggest the potential for substantial additional rare earth resources.⁷ This provides a significant upside optionality and a potential extension to the mine life of Longonjo's processing infrastructure, ensuring Pensana's relevance in the rare earth market for decades beyond the initial project scope. This combination offers investors exposure to a company with a tangible, de-risked near-term production profile, coupled with the high-reward potential of a major new discovery.

Attribution of Recent Price Spikes

Recent price movements in Pensana's stock can be attributed to specific project milestones and market developments. The most significant recent price spike, a **29% surge to 36.00 pence, occurred on March 18, 2025**, directly following the announcement that Pensana had secured full financing approvals for the Longonjo rare earth project, totaling approximately US\$268 million.²⁰ This event was a major de-risking catalyst, confirming the financial viability of the primary project.

More recently, the July 28, 2025, announcement regarding "Longonjo construction budget on track and Coola carbonatite drilling"³ did not result in a price spike; instead, the stock was down 9.2%.¹⁰ This suggests that while the news was fundamentally positive, the market may have already factored in these construction updates, or other broader market dynamics and recent equity issuance¹² influenced the short-term price action.

Company De-risking

The combined projects significantly de-risk Pensana's overall risk profile and enhance its upside potential, creating a unique investment proposition often described as having the "risk of a near-term producer, but the upside potential of a junior explorer."

- **Financial De-risking:** The securing of US *268millioninfinancingforLongonjo[20, 21]substantiallyreducethefundingriskfortheprimaryproject.Thephasedcapexapproach, withUS* 105 million deferred, further aids financial management.⁷
- **Execution De-risking:** Confirmation that Longonjo construction is on track and within budget ¹⁰, along with the mobilization of key engineering partners, mitigates construction and operational ramp-up risks.
- **Market De-risking:** Strategic offtake agreements with major players like Toyota Tsusho and ReElement Technologies ¹¹ secure market access for Longonjo's MREC, reducing commodity price exposure and sales risk.
- **Resource Longevity & Diversification:** The Coola project, with its potential for significant additional rare earth mineralization, acts as a long-term resource hedge. It offers the potential to extend the life of the Longonjo processing infrastructure ¹⁰, thereby diversifying the resource base and reducing reliance on a single deposit. This geological optionality provides a buffer against unforeseen challenges at Longonjo.
- **ESG & Strategic Alignment:** Pensana's strong commitment to sustainable practices, including low-carbon power and reagent recycling ²⁴, aligns with global ESG mandates. Its role in establishing an independent, non-Chinese rare earth supply chain positions it favorably amidst increasing geopolitical focus on critical mineral security.¹⁵ This strategic importance can attract government support and strategic partnerships, further de-risking the enterprise.

Overall Analogy/Summary

Pensana can be viewed as a "**two-engine aircraft**" navigating the critical minerals landscape. The **Longonjo project serves as the primary engine**, providing the immediate thrust towards production and cash flow, representing the company's tangible, de-risked pathway to becoming a significant rare earth producer. Its construction progress, secured financing, and offtake agreements are the fuel and mechanics ensuring this engine's reliable operation.

The **Coola project acts as the powerful secondary engine or booster**, currently in its testing phase. While not yet contributing to immediate flight, its immense, unquantified potential offers significant additional power for long-term endurance and accelerated growth. This secondary engine provides a critical layer of redundancy and expansion capability, shielding the company against unforeseen headwinds that might affect the primary engine and ensuring a prolonged trajectory in the high-demand rare earth market. Together, these two engines provide Pensana with a robust and resilient flight plan in the global energy transition.

Next Major Catalysts (Consolidated)

The following are the next big movers or catalysts expected for Pensana:

- **Longonjo Project:**
 - Continued construction progress and adherence to the US\$217 million capital cost schedule.¹⁰
 - Successful commissioning and commencement of initial production, anticipated in late 2026.¹⁸
 - Successful ramp-up to full Phase 1 production (20,000 tpa MREC) and subsequent expansion to Phase 2 (40,000 tpa MREC).¹⁸
 - Further updates on the Saltend refinery's development and its integration with Longonjo's output.
- **Coola Exploration Project:**
 - Release of further definitive drill assays from the ongoing drilling program.¹⁰
 - Publication of a maiden Mineral Resource Estimate (MRE), which would quantify the resource potential and allow for more formal economic assessments.⁴⁵
 - Metallurgical test work results confirming the processability of Coola's mineralization and its suitability as a feedstock for Longonjo.⁴⁰
- **Strategic Partnerships & Offtake:**
 - Securing additional strategic offtake agreements for Longonjo's MREC and Saltend's separated products, further de-risking future revenue streams.
 - Updates on collaborations, such as the MOU with ReElement Technologies, to strengthen the rare earth value chain.¹³

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